

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 31)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

01 . 1

Describe coal gasification and coal liquefaction and explain how each increases the usefulness of coal.

[4 marks]

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01 . 2

Describe what methane hydrate is and where it is found.

[2 marks]

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01 . 3

Describe **two** proposed methods of extracting methane from hydrate deposits and explain **two** environmental risks.

[4 marks]

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10

0 2

A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha ⁻¹	Carbon in soil dead organic matter / t ha ⁻¹
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 2

. 1

Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer

t/ha

0 2

. 2

Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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0 2

. 3

Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

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2

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5

03

Figure 1 shows the daily deaths in London and the mean concentration of smoke in the air during the smog of December 1952.

Figure 1

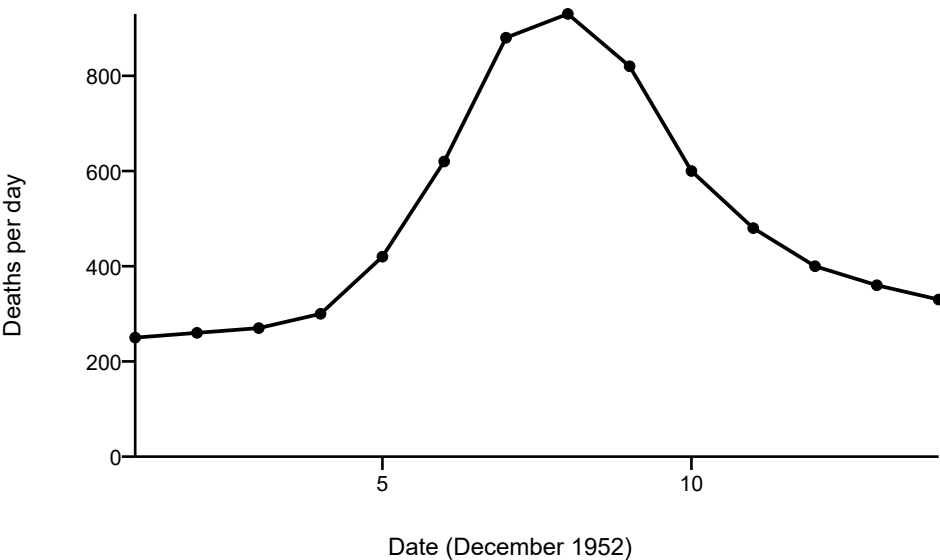


Table 1

Date	3 Dec	5 Dec	7 Dec	8 Dec	10 Dec	12 Dec
Smoke / mg m ⁻³	0.4	1.5	4.4	4.5	1.2	0.5

03

1

Describe the relationship between smoke concentration and daily deaths shown in Figure 1 and Table 1.

[2 marks]

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03

2

Calculate the percentage increase in daily deaths between 1 December and 8 December.

[2 marks]

Answer

%

03 . 3

Explain how the weather conditions in December 1952 caused the smog to form.

[3 marks]

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03 . 4

Explain why most of the deaths were among the elderly, the very young and people with existing respiratory disease.

[2 marks]

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03 . 5

State the legislation introduced in the UK in response to this smog.

[1 mark]

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10

04

Table 1 shows measurements taken from a stream above and below the point where drainage water from an old metal-mine spoil heap enters it.

Table 1

Site	pH	Dissolved zinc / mg dm ⁻³	Dissolved oxygen / % saturation	Number of invertebrate taxa	Mayfly larvae present?
Upstream	7.1	0.02	96	18	Yes
50 m downstream	3.4	6.5	94	3	No
2 km downstream	5.9	1.1	95	9	No

04 . 1

Explain how the acidic drainage water is produced in the spoil heap.

[3 marks]

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04 . 2

Use Table 1 to explain why the number of invertebrate taxa falls so sharply 50 m downstream, even though dissolved oxygen is unchanged.

[3 marks]

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04 . 3

Suggest why the pH and zinc concentration have partly recovered 2 km downstream.

[2 marks]

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04 . 4

Describe how acid mine drainage can be controlled.

[2 marks]

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10

0 5 Table 1 shows the bypass ratio and first year of use of jet engines used on four airliners.

Table 1

Engine	Aircraft	Bypass ratio	First use
RR Olympus	Concorde	0 : 1	Early 1950s
RR Conway	Boeing 707	0.25 : 1	Late 1950s
P&W JT9D	Boeing 747	5 : 1	1969
RR Trent 1000	Boeing 787	10 : 1	2011

0 5 . **1** Explain how a high bypass ratio reduces engine noise.

[3 marks]

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0 5 . **2** Describe **two** other features of aircraft design that reduce noise.

[2 marks]

1

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2

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0 5 . **3** Explain how each of the following airport operating procedures reduces noise nuisance.

[4 marks]

Constant descent angle

Night flight restrictions

Noise-based landing charges / quota count

Multiple runways

0 5 . 4

Explain why aircraft are noisier on take-off than on landing.

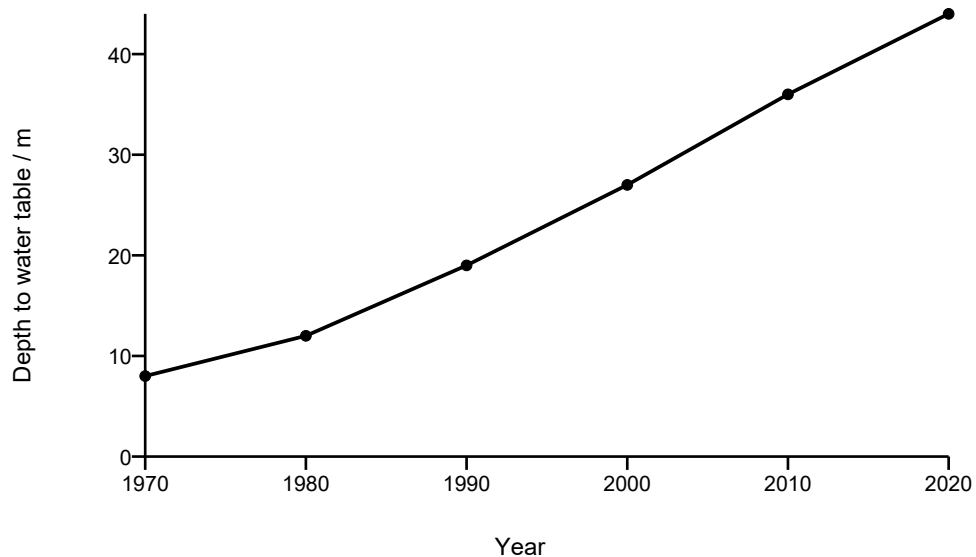
[1 mark]

10

0 6

Figure 1 shows the depth of the water table below the ground surface in an aquifer beneath a coastal farming region between 1970 and 2020.

Figure 1



0 6 . 1

Calculate the mean rate at which the water table fell between 1970 and 2020.

[1 mark]

Answer

m/yr

06

2

Explain why the water table is falling.

[2 marks]

06

3

Explain the features a rock must have to form an aquifer.

[3 marks]

06

4

Describe **four** consequences of over-exploiting this coastal aquifer.

[4 marks]

1

2

3

4

10

07

Table 1 shows the running mean of moth catches as a student increased the number of nights of light-trapping at one site.

Table 1

Number of nights	1	2	4	6	8	10	12	14
Running mean catch per night	31	18	24	21	22	22	21	22

07

1

Use Table 1 to suggest the minimum number of nights needed to obtain a reliable mean. Explain your answer.

[2 marks]

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07

2

Explain why moth surveys must be carried out at standardised times and weather conditions.

[3 marks]

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07 . 3

Distinguish between **accuracy** and **precision** of measurements.

[2 marks]

07 . 4

Explain the importance of a null hypothesis and standardised techniques in a scientific investigation.

[3 marks]

10

08 . 1

Explain why mineral extraction causes more land-use conflict than most other human activities.

[2 marks]

08 . 2

Describe how each of the following environmental impacts of an operating mine can be reduced.

[5 marks]

Dust

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Noise

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Turbid drainage water

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Acidic leachate from spoil

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Loss of amenity

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08 . 3

Explain how spoil heaps can be made stable and describe what happened at Aberfan in 1966.

[3 marks]

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10

09 . 1

Explain how ionising radiation damages living cells.

[3 marks]

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09 . 2

Distinguish between acute and chronic effects of radiation exposure.

[2 marks]

09 . 3

Distinguish between **exposure** and **contamination**.

[2 marks]

09 . 4

A worker receives an absorbed dose of 0.002 Gy from alpha radiation.
Calculate the effective dose in sieverts.

[2 marks]

Answer

Sv

09 . 5

State what is measured by the becquerel.

[1 mark]

10

Neonicotinoids are the most widely used group of insecticides. Explain how the properties of neonicotinoids affect their impact on bee populations and describe the methods used to reduce this impact.

ESP/7447/1/SET31

Extra space

10 . **2** State **one** way in which pesticide use in agriculture may be reduced by using natural processes.

[1 mark]

10

Write an essay on one of the following topics.

1	1	.	1
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To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?

[25 marks]

OR

1 1 . 2

Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

[illegible]

Do not write outside the box

END OF QUESTIONS